

Using Automatic Item Selection in Multi-Informant Assessment of Child Mental Health

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This work was preregistered on October 1st, 2025 on OSF (osf.io/hzus8). The codebase for this study is publicly available on GitHub (<https://github.com/jlschnatz/multi-informant-aseba>). The authors have no conflicts of interest to disclose.

Abstract

“Comprehensive clinical assessment of mental health in children and adolescents often relies on multiple informants, such as parents, teachers and children themselves. However, discrepancies in reports from different raters are common, leading to challenges in interpreting the results of assessment procedures. These discrepancies may arise because some items capture behaviors that are commonly observed across informants, while others reflect informant-specific perspectives unique to each rater. At the same time, there is a need for economic instruments that reduce burden for respondents and clinicians while maintaining reliability and validity. To address these challenges the present study aims to develop a subtest containing 32 items of the commonly employed multi-informant assessment instrument Achenbach System of Empirically Based Assessment (ASEBA) that balances shared and informant-specific perspectives while retaining a manageable number of items for economic administration in clinical settings. This will be achieved within a structural equation modeling framework leveraging an automatic item selection approach to identify an item set that optimally satisfies a predefined set of construction criteria. The present study will use two waves of data of parent and child ratings on the ASEBA Child Behavior Checklist (CBCL/6-18) and Youth Self Report (YSR/11-18) from the SAFE children study (Tolan et al., 2016), a longitudinal randomized controlled trial with $N = 424$ children and their primary caregivers that participated in a family-centered preventive intervention program”

Keywords: Multi-Informant Assessment, Automatic Item Selection, Scale Construction, Achenbach System of Empirically Based Assessment

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Evidence-based clinical assessment in children and adolescents often relies on multiple informants, such as parents, teachers, and children themselves, to obtain a comprehensive understanding of child's mental health (Hunsley & Mash, 2007; Mash & Hunsley, 2005). This typically involves using parallel standardized questionnaires such as the widely used Achenbach System of Empirically Based Assessment (ASEBA) that includes self-report, parent-report, and teacher-report forms (T. M. Achenbach & Rescorla, 2001).

However, extensive meta-analytical empirical evidence has shown that discrepancies in reports from different informants (*informant discrepancies*) are common in these instruments (T. M. Achenbach et al., 1987; De Los Reyes et al., 2015; Jungersen, 2023), leading to challenges in understanding and interpreting the results of the assessment procedures.

Historically, informant discrepancies have been conceived as measurement error, treating them as interchangeable and informant-specific variance as nuisance (De Los Reyes et al., 2023). This line of interpretation stems from the seminal work of Campbell and Fiske (1959) on multi-trait multi-method analysis, which has shaped both theoretical and empirical efforts to emphasize common variance across informants (De Los Reyes & and Epkins, 2023).

More recent theoretical advances concerning the *Operations Triad Model* (De Los Reyes et al., 2013) have shifted the focus towards a more nuanced understanding of multi-informant assessment. Specifically, *converging operations* in this model refer to measurement conditions in which informants' reports align, leading to similar conclusions. The Operations Triad Model distinguishes among informant discrepancies based on their underlying sources. *Diverging* operations capture instances where informants offer distinct yet domain-relevant perspectives, whereas *compensating* operations reflect discrepancies that stem from measurement-related confounds, such as random error or rater biases (De Los Reyes & and Epkins, 2023). This shift in perspective has led to a growing body of literature that emphasizes the importance of informant-specific perspectives in multi-informant assessment (De Los Reyes, 2011; De Los Reyes et al., 2023).

This reconceptualization has contributed to a paradigm shift in measurement, highlighting the need to account for both common variance and meaningful informant-specific variance (De Los Reyes & and Epkins, 2023). However, because most existing instruments were developed with an emphasis on common variance, they may lack sensitivity to detect domain-relevant informant discrepancies, highlighting the need to refine current measures or develop new tools that explicitly integrate both shared and informant-specific item content (De Los Reyes & and Epkins, 2023).

In addition to the need for instruments that capture both shared and informant-specific perspectives, there is also a growing demand for shortened tests that are feasible to implement in time-limited clinical settings (Goetz et al., 2013). For example, within the widely used ASEBA system introduced above, the Brief Problem Monitor (BPM) has been developed as a shortened test for the economical assessment of emotional and behavioral problems (T. Achenbach et al., 2011). However, the BPM uses the same items across all informants, which may not be optimal for capturing meaningful informant-specific perspectives. This highlights the need for instruments that can effectively balance shared and informant-specific perspectives while also being economical for use in clinical settings.

Objectives and Research Questions

The primary objective of this study is to develop a multi-informant subtest of the Achenbach System of Empirically Based Assessment (ASEBA) that balances shared and informant-specific perspectives, while also retaining a manageable number of items for economic administration in clinical settings.

Specifically, we aim to identify a combination of items that function invariantly across informants (capturing common variance) and items that exhibit informant-specific functioning (capturing unique variance relevant to each informant). As part of our construction goal, we will attempt to ensure that all ASEBA clinical subscales are reflected in the final item set to additionally support the content validity of the subtest. To achieve this objective, we will leverage the combinatorial nature of item selection (Schultze, 2017) using an automatic item

selection approach to identify item sets that optimally satisfy a predefined set of construction criteria.

Hypotheses

The following hypotheses will be tested in the current study:

1. H_1 : The model fit of the developed subtest will meet a predefined set of Minimum Viable Criteria (MVC) (see section #link()[Minimum Viable Criteria (MVC)] for details relating to their definition).
2. H_2 : The subset of items intended to reflect the shared cross-informant perspective will demonstrate measurement invariance across different informants (child and parent).
3. H_3 : The subset of items intended to reflect the informant-specific perspectives will demonstrate informant-specific functioning, indicating that these items capture unique variance relevant to each informant's perspective.

Methods

Table 1

Cross-Informant Agreement and Reliability Estimates for all ASEBA Subscales

Sub-scale	Agreement	CBCL			YSR		
		α	α^*	N	α	α^*	N
AB	0.52	0.94	0.635	18	0.90	0.514	17
AD	0.45	0.84	0.447	13	0.84	0.447	13
AP	0.48	0.86	0.551	10	0.79	0.455	9
RB	0.55	0.85	0.400	17	0.81	0.362	15
SP	0.49	0.82	0.453	11	0.74	0.341	11
SC	0.40	0.78	0.392	11	0.80	0.444	10
TP	0.37	0.78	0.321	15	0.78	0.371	12
WD	0.40	0.80	0.500	8	0.71	0.380	8

Note. Test

Sample**Table 2***Descriptive Statistics of the Category Levels for the CBCL and YSR in Both Samples*

Subscale	Training			Testing		
	0	1	2	0	1	2
CBCL						
AB	62.9 (10.8)	13.9 (9)	2.7 (2.1)	63.8 (13.4)	13.8 (8.6)	2.4 (1.9)
AD	70.2 (12.1)	9.9 (6.4)	1.5 (1)	73.3 (15.5)	10.2 (6.4)	1.4 (0.8)
AP	63.6 (14.5)	16.1 (6.6)	2.6 (1.7)	60.9 (8.5)	15 (7.4)	2.6 (1.4)
RB	72.1 (15.2)	8.1 (7.9)	2.7 (1.8)	74.1 (18.7)	8.9 (8.6)	2.2 (1.3)
SC	72.8 (10.5)	8.1 (4.5)	1.1 (0.9)	71.8 (10.5)	8 (4)	1.1 (1)
SP	68.8 (6.9)	9.2 (5.7)	1.5 (1.5)	73.4 (12.8)	8.8 (5.2)	1.3 (1.7)
TP	82.1 (25)	6 (5.6)	1.2 (1.1)	79.6 (16)	5.6 (4.8)	1.1 (1.3)
WD	60.1 (9.8)	16.2 (7.9)	3.2 (2)	59.2 (10.9)	16.7 (9.5)	2.5 (1.6)
YSR						
AB	57.9 (11.3)	15.7 (8)	4.4 (4)	58.1 (11.1)	14.8 (7.9)	3.8 (3.9)
AD	63.4 (11.2)	11.4 (8.5)	3.1 (3)	61.9 (11.9)	11.7 (9.1)	3.1 (3.2)
AP	51.5 (9.6)	21.3 (6.6)	5.2 (3.4)	52.4 (10.2)	19.6 (6.9)	4.8 (3.4)
RB	60.8 (13.9)	13.7 (10.9)	3.5 (3.5)	58.5 (13.9)	14 (10.4)	4.2 (4.1)
SC	62.4 (6.8)	13.1 (5.6)	2.5 (1.6)	60.5 (7.1)	13.5 (5.6)	2.8 (1.9)
SP	65 (5.4)	10.9 (4.8)	2.1 (0.8)	64.8 (4.7)	10.3 (4.1)	1.6 (0.8)
TP	61 (16.6)	15.4 (8.6)	4.3 (2.7)	60.2 (17.1)	14.9 (9.2)	4.4 (2.7)
WD	53.8 (10.5)	19.5 (7)	4.6 (4)	54.2 (10.4)	17.6 (6.7)	4.9 (4)

Note. 0: Not true, 1: Somewhat or sometimes true, 2: Very true or often true. Values before parentheses indicate the mean proportion across all items belonging to the subscale for each each. Values in parentheses indicates the respective standard deviations. WD: Withdrawn/Depressed, TP: Thought Problems, SP: Social Problems, SC: Somatic Complaints, RB: Rule-breaking Behavior, AP: Attention Problems, AD: Anxious/Depressed, AB: Aggressive Behavior. CBCL: Child Behavior Checklist, YSR: Youth Self Report.

Objective Function**Criteria**

To select a set of items that meets the construction goal, an objective function will be defined, consisting of the following criteria:

Model Fit.

Two model fit indices will be included as criteria for the objective function: The sample-corrected robust Root Mean Square Error of Approximation (Li & Bentler, 2006).

$$\text{RMSEA} = \sqrt{\max\left\{0, \frac{\hat{F}_{\text{ML}}}{df} - \frac{\hat{c}}{n-1}\right\}}$$

where $\hat{c}$ is correction constant, $\hat{F}_{\text{ML}}$ is the value of the minimized fit function, df are the degrees of freedom and n is the sample size (Brosseau-Liard et al., 2012). This corresponds to the “rmsea.robust” fit measure in the *lavaan* package (Rosseel, 2012).

Additionally, the Standardized Root Mean Square Residual (Bentler, 1995) will be included as a second model fit criterion

$$\text{SRMR} = \sqrt{\frac{2}{p(p+1)} \sum_{u=1}^p \sum_{v=1}^p \left(\frac{s_{uv} - \hat{\sigma}_{uv}}{s_{uu} s_{vv}} \right)^2}$$

where p is the number of manifest variables in the model s_{uv} is the sample covariance between manifest variables u and v and $\hat{\sigma}_{uv}$ is their estimated covariance. This corresponds to the “srmr” criterion in the *lavaan* package (Rosseel, 2012).

Reliability.

We will additionally include a criterion for the average reliability across the four latent factors, given by

$$\bar{\omega} = \frac{1}{l} \sum_{j=1}^l \left[\frac{\left(\sum_{i=1}^p \lambda_{ijkl} \right)^2}{\left(\sum_{i=1}^p \lambda_{ijkl} \right)^2 + \sum_{i=1}^p \theta_{ijkl}} \right]$$

where l is the number of latent factors (in this case $l = 4$), λ_{ijkl} is the factor loading of item i ($i = 1, \dots, I_j$) within the j -th clinical subscale ($j = 1, \dots, J$) for the k -th information ($k \in \{1, 2\}$) loading on the l -th latent factor ($l \in \{\text{CI} - \text{C}, \text{CI} - \text{P}, \text{IS} - \text{C}, \text{IS} - \text{P}\}$), and θ_{ijkl} is the corresponding error variance of that item.

Cross-Informant Regression.

Highlighted in orange in the model in Figure 1, the regression coefficient γ from the cross-informant parent factor (CI-P) to the cross-informant child factor (CI-C) will be included as

a criterion in the objective function. This coefficient reflects the extent item selected for these factors are answered consistently across both informants.

Informant-Specific Correlation.

Finally, the latent correlation φ highlighted in blue in Figure 1 between the informant-specific factors (IS-C and IS-P) will be included as a criterion in the objective function. This correlation reflects the extent to which the items selected for these factors capture informant-specific perspectives by aiming for a low correlation between the two factors.

Combination of the Criteria

Each criterion will be transformed onto a common scale using the cumulative distribution function of the normal distribution $\Phi(x_q; \hat{\mu}_{x_q}, \hat{\sigma}_{q_w})$, with parameters corresponding to difficulty $\hat{\mu}_{x_q}$ (point of maximum discrimination) and discrimination $\hat{\sigma}_{q_w}$ (slope). These parameters will be empirically estimated from $k = 5000$ random solutions sampled from the search space of all possible item combinations from the model specified in Figure 1. The mean and standard deviation of the top 10% of valid solutions will be used for the transformation.

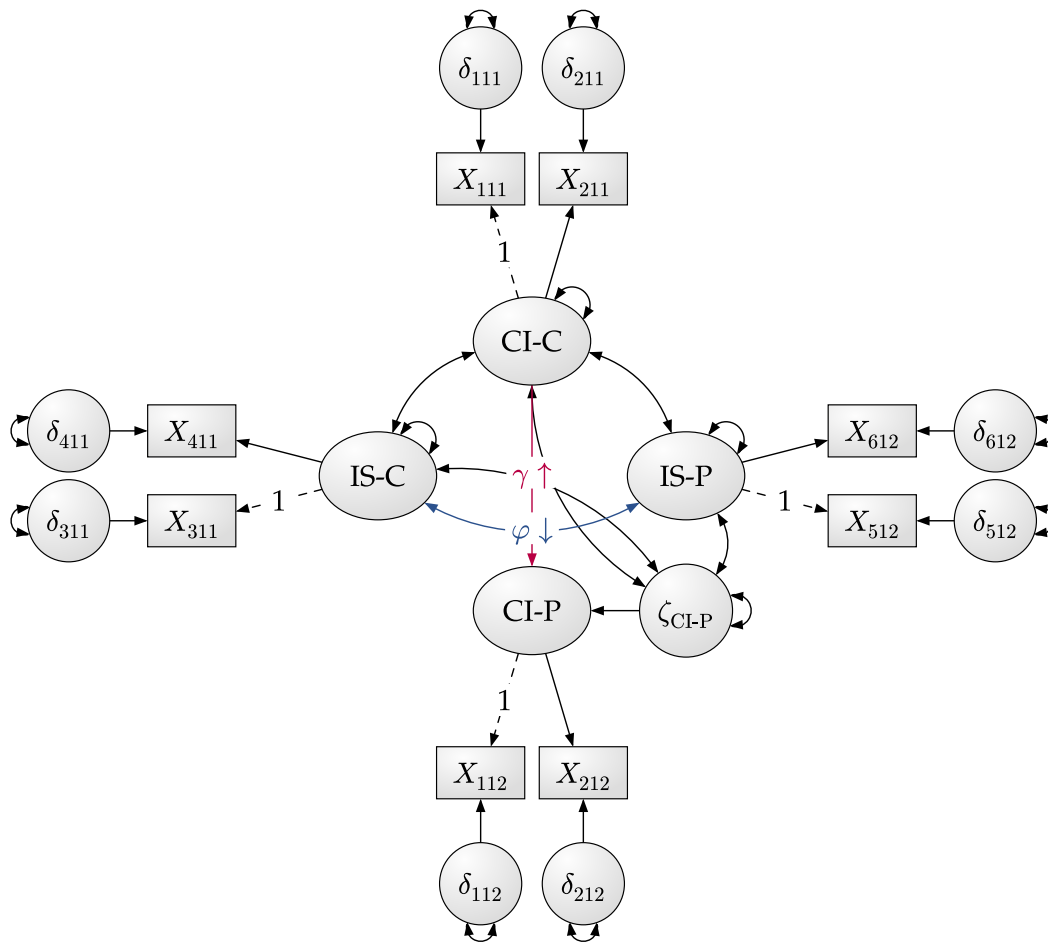
The combined empirical objective function f is then given by a weighted sum of the set of transformed criteria $\mathcal{X} : \{\text{RMSEA}, \text{SRMR}, \hat{\omega}, \varphi, \gamma\}$:

$$f = \sum_{q=1}^r w_q \cdot \Phi(x_q; \hat{\mu}_{x_q}, \hat{\sigma}_{q_w})$$

with weights $\mathcal{W}$ set as $w_{\text{RMSEA}} = w_{\text{SRMR}} = w_{\gamma} = w_{\varphi} = \frac{1}{6}$ and $w_{\hat{\omega}} = \frac{1}{3}$. This means that objective function balances equally between model fit criteria, criteria relating to the multi-informant quality of the instrument, and reliability.

Figure 1

Hypothesized CTC(M-1) Structural Equation Model of a Single Clinical Subscale of ASEBA



Note. Subscripts X_{ijk} denote the item i , clinical subscale j and rater k . Double-headed arrows represent covariances, single-headed arrows represent regression. Round δ nodes are error variables of each item. CI-C: Cross-Informant Child factor. CI-P: Cross-Informant Parent factor, IS-C: Informant-Specific Child factor, IS-P: Informant-Specific Parent factor. φ : latent correlation between the informant-specific factors. γ : latent regression parameter of the cross-informant factors. Figure generated using the typst package *fletcher* (Wilson, 2026).

Results

Table 3

Maximum Pheromone, Convergence and Number of Combinations of the Brute-Force Search

Subscale	$\max(\varphi)$	Viable Solutions (%)	N
AB	0.85	8.23	2829888
AD	0.88	11.58	474552
AP	0.83	6.47	58320
RB	0.87	3.20	1499400
SC	0.89	5.02	111375
SP	0.95	4.60	166375
TP	0.86	3.30	317625
WD	0.82	3.30	21952

Note. WD: Withdrawn/Depressed, TP: Thought Problems, SP: Social Problems, SC: Somatic Complaints, RB: Rule-breaking Behavior, AP: Attention Problems, AD: Anxious/Depressed, AB: Aggressive Behavior. RMSEA: Root Mean Squared Error of Approximation

Table 4

Estimated Means and Standard Deviations of Random Solutions for Objective Function Parameters

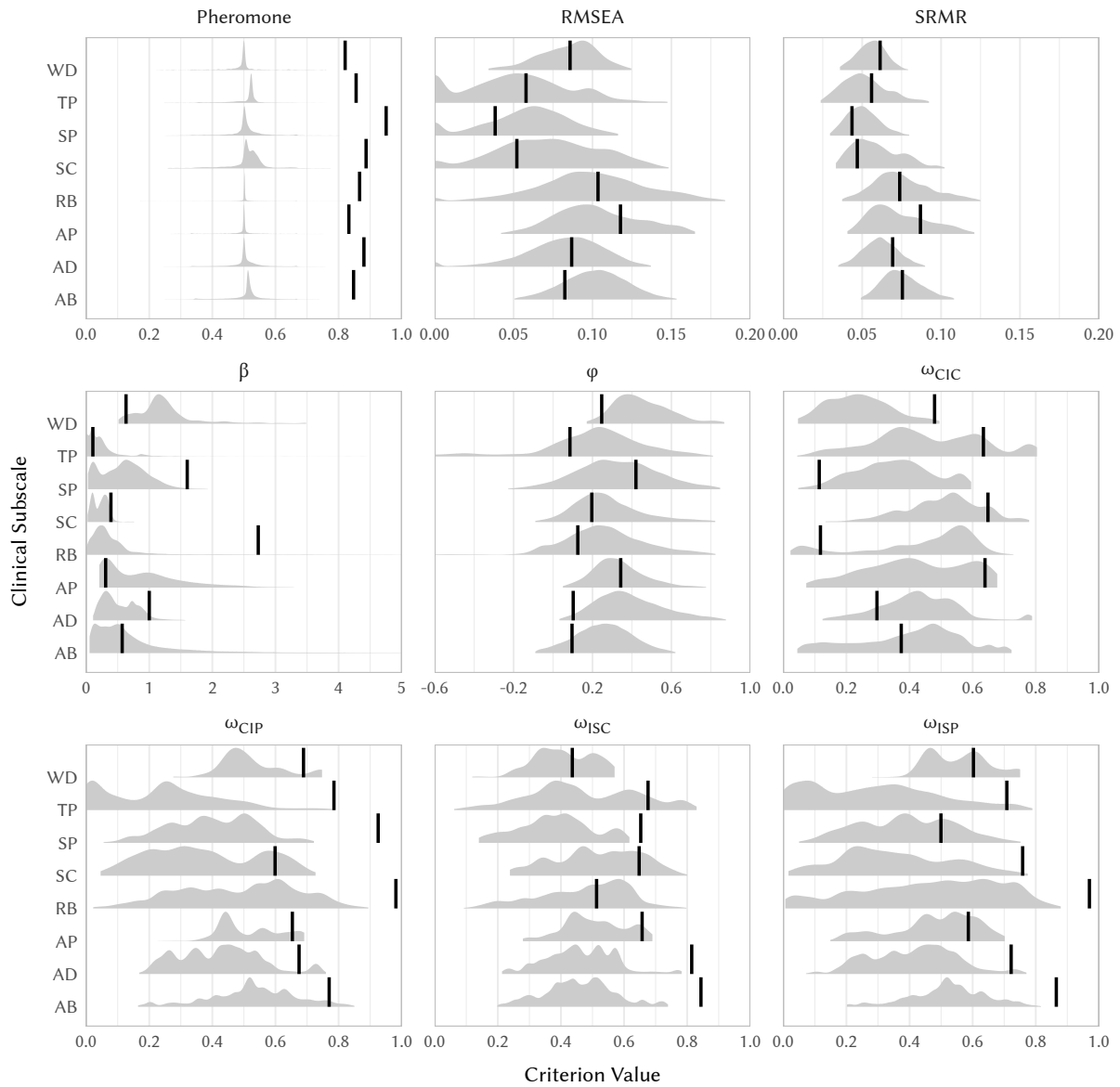
Subscale	RMSEA	SRMR	ω	γ	φ
AB	0.14 (0.01)	0.10 (0.01)	0.61 (0.03)	4.08 (2.59)	0.50 (0.05)
AD	0.12 (0.01)	0.08 (0.01)	0.55 (0.04)	1.10 (0.24)	0.75 (0.08)
AP	0.15 (0.01)	0.11 (0.01)	0.58 (0.02)	2.58 (0.54)	0.61 (0.08)
RB	0.16 (0.01)	0.11 (0.00)	0.59 (0.02)	3.91 (1.54)	0.72 (0.12)
SC	0.13 (0.01)	0.09 (0.01)	0.57 (0.03)	0.49 (0.22)	0.70 (0.10)
SP	0.10 (0.01)	0.07 (0.01)	0.49 (0.03)	1.36 (0.35)	0.75 (0.07)
TP	0.12 (0.02)	0.08 (0.01)	0.55 (0.04)	3.14 (2.77)	0.60 (0.10)
WD	0.11 (0.01)	0.07 (0.00)	0.51 (0.02)	2.97 (0.51)	0.81 (0.08)

Note. Values before parentheses indicate mean, values in parentheses standard deviation. WD: Withdrawn/Depressed, TP: Thought Problems, SP: Social Problems, SC: Somatic Complaints, RB: Rule-breaking Behavior, AP: Attention Problems, AD: Anxious/Depressed, AB: Aggressive Behavior. RMSEA: Root Mean Squared Error of Approximation, SRMR: Standardized Root Mean Residual, ω : Reliability, φ : latent correlation between the informant-specific factors of the child and parent informants, γ : regression weight of the cross-informant factors between the

child and parent informants. These values are determined based on the top 10% of $k = 5000$ random solution

Figure 2

Distribution of Empirical Pheromone and Objective Function Criteria Across Clinical Subscales

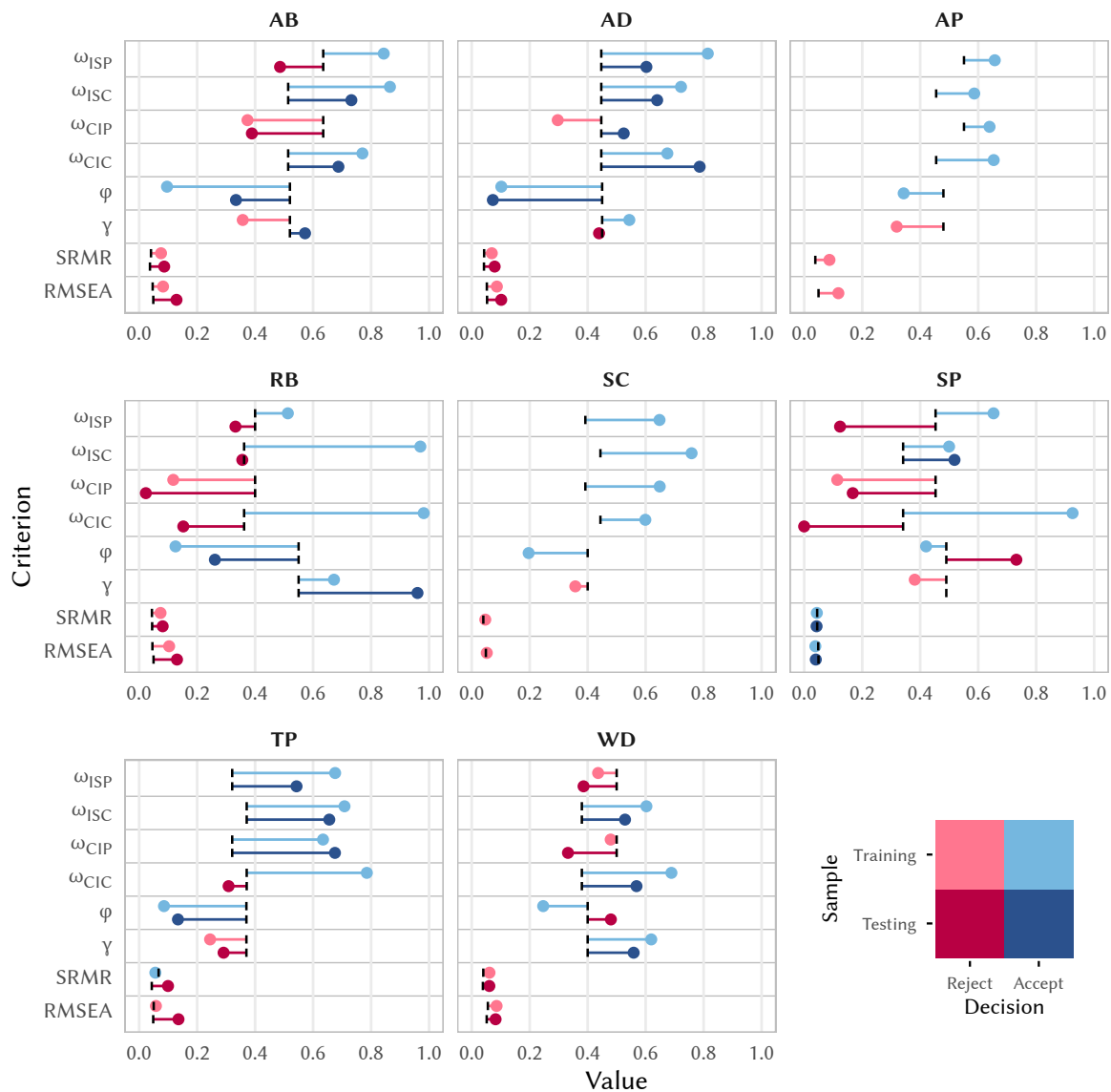


Note. WD: Withdrawn/Depressed, TP: Thought Problems, SP: Social Problems, SC: Somatic Complaints, RB: Rule-breaking Behavior, AP: Attention Problems, AD: Anxious/Depressed, AB: Aggressive Behavior. RMSEA: Root Mean Squared Error of Approximation, SRMR: Standardized Root Mean Residual, ω : Reliability of the latent factors cross-informant and informant specific child and parent factors, γ : Regression weight of the cross-informant factors between

the child and parent informants, ϕ : latent correlation between the informant-specific factors of the child and parent informants. Black bar indicates criterion value, where the specific item combination yields the maximum pheromone values.

Figure 3

Brute-Force Identified Criteria Reject Cutoffs Across Subscales in Training and Testing Samples



Note. WD: Withdrawn/Depressed, TP: Thought Problems, SP: Social Problems, SC: Somatic Complaints, RB: Rule-breaking Behavior, AP: Attention Problems, AD: Anxious/Depressed, AB: Aggressive Behavior. RMSEA: Root Mean Squared Error of Approximation, SRMR: Standardized Root Mean Residual, ω : Reliability of the latent factors cross-informant and informant

specific child and parent factors, γ : Regression weight of the cross-informant factors between the child and parent informants, ϕ : latent correlation between the informant-specific factors of the child and parent informants. Black bars represent the predefined cutoff values for each criterion. Blue and orange symbols indicate the decision outcome (accept vs. reject), with color saturation differentiating the sample type (darker for training, lighter for testing). No testing-sample results are reported for the SC and AP subscales due to model nonconvergence. For the SP subscale, the β parameter is omitted because of negative estimated latent factor variances.

Discussion

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